

🗨️ Advanced Machine Learning — From Scratch

14 ML projects implemented from the ground up using only NumPy — no PyTorch, no TensorFlow (unless explicitly noted).

Every algorithm, every gradient, every matrix multiplication — written by hand.

Based on the **Advanced Machine Learning** course at [TU Hamburg](#) (Prof. Zemke, WS 2025/26).
This repository proves deep understanding of ML fundamentals — not just `model.fit()`.

👤 About Me

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Passionate about AI, Deep Learning, and building things from scratch.

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📖 Project Roadmap

🟢 Stage 1 — Neural Network Fundamentals (Lectures 1-3)

#	Project	Key Concepts	Highlights
01	<u>Feedforward Net from Scratch</u>	Forward pass, backpropagation, minibatch SGD, Adam	97% MNIST accuracy with pure NumPy
02	<u>Activation Function Explorer</u>	9 activations, vanishing gradient, real benchmark	Interactive explorer + gradient flow analysis
03	<u>SVM Classifier + Kernel Trick</u>	Hyperplanes, margin, RBF/polynomial kernels, SMO	100% on XOR with RBF kernel
04	<u>Optimizer Showdown</u>	SGD, Momentum, NAG, Adagrad, RMSprop, Adam, Nadam	7 optimizers racing on 5 test functions
05	<u>Weight Initialization Study</u>	He, Xavier, Orthogonal, signal propagation	Zeros = 50%, He = 98.6% on same network

● Stage 2 — CNNs & Computer Vision (Lectures 4-7)

#	Project	Key Concepts	Highlights
06	<u>CNN from Scratch (im2col)</u>	Conv2D, MaxPool, im2col + GEMM, backprop through CNN	98.3% accuracy, im2col like PyTorch internally
07	<u>Image Filter Lab</u>	Sobel, Gaussian, Laplacian, Sharpen, 14 filters total	Edge detection, denoising, filter composition
08	<u>LeNet-5 on MNIST</u>	Original (1998) vs Modern CNN, BatchNorm, Dropout	~99% accuracy, CNN history timeline
09	<u>Fast Convolution Benchmark</u>	Naive vs im2col vs FFT vs Winograd vs Separable	im2col 50× faster, FFT wins for large kernels
10	<u>Adversarial Attack Demo</u>	FGSM, Targeted FGSM, I-FGSM, adversarial training	Accuracy drops from 92% → 0% with $\epsilon=0.25$

● Stage 3 — Sequences & Generative AI (Lectures 8-13)

#	Project	Key Concepts	Highlights
11	<u>RNN Text Generator</u>	Elman network, BPTT, hidden states, temperature sampling	Character-level text generation from scratch
12	<u>VAE Image Generator</u>	Encoder, decoder, reparametrization trick, KL divergence	2D latent space + interpolation between digits
13	<u>Transformer from Scratch</u>	Self-attention, multi-head attention, positional encoding	Full encoder with attention visualization
14	<u>Simple Chatbot (Fine-tuning)</u>	Pretrain → Fine-tune → Chat, transfer learning	Same pipeline as GPT/ChatGPT at mini scale

▢ Mathematical Foundations Covered

This repository implements the core mathematics from all 13 lectures:

Topic	Formula	Project
Feedforward evaluation	$a_1 := x, z_i := W_i a_i + b_i, a_{i+1} := \sigma(z_i)$	01
Backpropagation	$\delta_i = \sigma'(z_i) \odot (W_{i+1}^T \delta_{i+1})$	01
SVM margin	$\text{margin} = 2/\ w\ _2$	03
Adam optimizer	$p \leftarrow p - \eta \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$	04
He initialization	$\sigma = \sqrt{2/n_{in}}$	05
2D convolution output	$o = \lfloor (i + 2p - k)/s \rfloor + 1$	06
im2col + GEMM	$\text{output} = W \cdot \text{im2col}(X) + b$	06, 09
FGSM attack	$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x L)$	10
RNN hidden state	$h_t = \tanh(W_{hh} h_{t-1} + W_{hx} x_t + b_h)$	11
VAE loss	$\mathcal{L} = \text{MSE} - \frac{1}{2} \sum (1 + \log \sigma^2 - \mu^2 - \sigma^2)$	12
Scaled attention	$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) \cdot V$	13
LLM pipeline	Pretrain $\rightarrow$ Fine-tune $\rightarrow$ RLHF	14

Tech Stack

- **Language:** Python 3.10+
- **Core:** NumPy (all from-scratch implementations)
- **Visualization:** Matplotlib
- **Optional:** TensorFlow/Keras (Project 08 only)
- **No frameworks for core ML:** Every forward pass, backward pass, optimizer, and loss function is hand-coded

Quick Start

```
bash
```

```
# Clone the repository
git clone https://github.com/YOUR_USERNAME/advanced-ml-from-scratch.git
cd advanced-ml-from-scratch

# Pick any project and run it
cd 01_feedforward_net
pip install -r requirements.txt
python train_xor.py      # Quick test (2 seconds)
python train_mnist.py    # Full training (~3 minutes)
```

Each project is self-contained with its own `README.md`, `requirements.txt`, and pre-generated figures.

Key Results at a Glance

Project	Task	Accuracy / Result
01 — FeedforwardNet	MNIST digits	~97%
03 — SVM	XOR (RBF kernel)	100%
06 — CNN	Digit classification	98.3%
08 — LeNet-5	MNIST (Keras)	~99%
09 — Fast Conv	im2col vs Naive	50× speedup
10 — FGSM	Attack success	92% → 0%
12 — VAE	Latent space	Smooth 2D manifold
13 — Transformer	Sentiment	71% (from scratch!)

Course Reference

All projects are based on:

Advanced Machine Learning — Prof. Jens-Peter M. Zemke
Institut für Mathematik, TU Hamburg, WS 2025/26

Lecture	Topic	Projects
1	Basics, SVM, Activations	01, 02, 03
2	Backpropagation, SGD	01, 04
3	Optimizers, Initialization, Regularization	04, 05
4	CNN: Convolution, Pooling	06, 07
5	CNN History, LeNet-5	08
6	FFT, im2col, Winograd	09
7	Understanding CNN, Adversarial Attacks	10
8	Recurrent Neural Networks	11
11	Autoencoder & GAN	12
12	Attention & Transformer	13
13	LLM, Finetuning, Fast Inference	14

License

MIT License — feel free to use, modify, and learn from this code.

Built with curiosity, NumPy, and a lot of matrix multiplications. 